SCI-ALIGN v0.1

Engineering Conformance Specification of the Shared Scientific Authority

Stage B Targeted Revision r0.3 — Author Draft

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Abstract

This engineering-facing specification contains the complete six-module SCI-ALIGN v0.1 formal authority: notation, definitions, assumptions, all twenty canonical equations, all fifty-five stable requirements, and all twenty-six stable predictions. The scientist-facing rationale is a separate narrative crosswalk to this complete contract. The title names an intended future comparison activity; this document neither inspected nor assesses an implementation and therefore makes no conformity, validation, freeze, readiness, or production claim.

Use rule. A future assessment may compare a named implementation and exact evidence set against these requirements only after scientific-owner disposition of the affected open decisions. A typed unavailable claim is not a failure of unrelated contract content and is never silently replaced by an implementation convention.

Contents

1 Notation and identity

The contract is observation-scoped. A quantity's unit, frame, topology, support, and missing-data meaning belong to that quantity and never follow from a shared container name. Observation identity is externally supplied; reduction-local observation, sample, slot, and internal scan indices are zero-based. Internal intervals are half-open.

Symbol or term	Identity and shape	Unit or domain	Meaning
o	stable observation identity	external identifier	Parent of every native occurrence, grid, slot, interval, and realized plan.
$i \in \{D, T, H\}$	detector, telescope, optional HWPR interface	interface identifier	Native interface; $i_{\mathrm{ref}} = D$ is the selected v0.1 reference interface.
d	detector occurrence or stable detector identity	detector identifier	Detector identity is not detector-column position; p is reserved for a map pixel.
k	ordered native row	zero-based integer	Native occurrence identity is bound to interface, observation, and row, not reconstructed from time alone.
E_i, u_{ik}	named epoch and native coordinate	seconds after checked conversion	Producer-owned event/epoch meaning and ordered native time coordinate.
t_{ik}^{ref}	corrected native event coordinate	detector-reference seconds	Checked epoch conversion plus the exactly-once admitted interface offset.
h_i, d_{ik}	cadence and native support/duration	seconds	Producer facts; neither is inferred from row count or nominal slots.
$[B_i^-, B_i^+)$	acquisition bounds	seconds	Half-open producer-declared native acquisition support, mapped by the same admitted clock relation.
$\delta_{i\to\mathrm{ref}}$	interface-offset record value	seconds	Positive when added to place the native coordinate later on detector-reference time; $\delta_{\mathrm{ref}\to\mathrm{ref}} = 0$.
ϕ_D, h_D	detector-grid phase and cadence	seconds	Define the nominal detector-reference lattice.
s	stable ALIGN detector-reference slot	zero-based integer	Stable ALIGN identity is (o, s), independent of any selected window or array layout.
j	local storage row	zero-based integer	A view-local row derived from s and the selected window; never an external identity and never a source for reconstructing s.
n	stable RTC output-sample identity	downstream identifier	Reserved for RTC output identity; it is not an ALIGN slot or ALIGN local row.

Symbol or term	Identity and shape	Unit or domain	Meaning
t_s, I_s	nominal ALIGN slot time and cell support	seconds	Nominal output relation; neither proves physical event time, integration, nor exposure.
ϵ_{slot}	assignment tolerance	seconds	Declared with $0 \leq \epsilon_{\text{slot}} < h_D/2$.
V_D	native paired detector-value container over rows and detectors	coordinate-specific declared units	Tune/readout-produced $(x, r)^{\text{acq}}$, with common occurrence relation but coordinate-specific validity and payload availability.
V_{Tv}, V_{Hv}	native values for named field v	declared unit and source frame	Telescope/observing-state or optional HWPR producer facts.
A_{ALIGN}	realized detector-pair temporal mapping	dimensionless	One mapping applied separately to x^{acq} and r^{acq}, never mixing the coordinates.
G_D, I_T, I_H	realized detector, telescope, and HWPR mappings	dimensionless	Conditional mappings after identities, operators, timing, support, and policy resolve.
S_q	processing-window selection	dimensionless	Selects a stable half-open window without changing slot or physical-scan identity.
A_q	realized block mapping for selected window q	dimensionless	Data-derived and conditioned on timestamps, producer state, offsets, and resolved plan.
$\mathcal{S}_{\text{ALIGN}}$	ALIGN product-state tuple	categorical and relational	Origin, local validity, method, reason, source identities, and weights; distinct from acquired microwave quadrature Q.
e_{sd}^{acq}	physical acquired detector exposure	seconds	Native physical integration support attributable to an original detector occurrence d at slot s, independent of payload scientific validity; zero for synthesized, missing, or unoccupied support.
e_{sd}^{vo}	valid-original detector exposure	seconds	Subset of e_{sd}^{acq} whose original payload and required ALIGN-local mapping facts are valid.
C_v, C_y	input and output covariance	squared declared units	Formal covariance when admitted inputs and the selected detail tier make it available.
R_q	conditional mapping response	output per template amplitude	ALIGN response conditional on the realized mapping, not a downstream end-to-end transfer function.

1.1 Typed state axes

The following axes are independent unless an explicit named rule composes them:
Origin
original, synthesized, or unavailable, refined without upgrading origin by original-invalid and guarded facts.

Method
exact, linear, circular, held, a separately authorized continuity-surrogate identity, or none.
Availability and validity
Presence, payload availability, finiteness, producer validity, ALIGN mapping validity, support, uncertainty availability, and reason are separately represented; no numeric sentinel is sufficient.
Five causes
Detector-occurrence identity, signal-coordinate validity, ALIGN mapping validity, AST coordinate validity, and downstream use-specific eligibility are non-substitutable causes. One cause neither aliases nor rescues another.
Exposure facts
Physical acquired exposure, valid-original exposure, synthesized/surrogate support, missing/unoccupied support, and later guarded, retained, or use-qualified exposure are separate facts. Later policy never rewrites whether a physical integration occurred.
Resolution state
Requested, effective, observation-resolved, and realized state flow one way. Lifecycle and generation provenance identify the owning request and observation.

The word *detector-reference* names the selected detector-stream reference interface, its admitted clock relation, and its assigned grid. It never names a reference detector.

2 Scientific definitions and ownership boundary

2.1 Purpose and ordinary signal order

SCI-ALIGN relates native detector, telescope, and admitted auxiliary-stream occurrences to one immutable, observation-scoped detector-reference sample identity before scientific conditioning. The ordinary detector-signal order is

$$(I,Q)^{\mathrm{acq}} \longrightarrow \text{Tune/readout} \longrightarrow (x,r)^{\mathrm{acq}} \longrightarrow \text{SCI-ALIGN} \longrightarrow (x,r)^{A}.$$

Tune/readout owns the native I,Q-to-x,r transformation, its units, normalization, and Tune/mapping-revision boundary. ALIGN starts from the exact native paired $(x,r)^{\mathrm{acq}}$ occurrences. It does not align I,Q, assume that Tune conversion commutes with interpolation or synthesis, mix x and r, or fill either coordinate from the other.

ALIGN owns checked epoch normalization and one exactly-once admitted offset, the detector-reference grid and assignment relation, per-field alignment, gap/guard/interval identity, origin and synthesis facts, ALIGN-local mapping validity and support, physical acquired and valid-original exposure, conditional response and uncertainty availability, and realized lineage. It does not calibrate, filter, estimate correlated components, construct or correct coordinates, select a WCS center, choose a MAP estimator, deposit samples into pixels, author a named-use eligibility policy, or interpret polarization.

2.2 Paired detector occurrence relation

One realized A_{ALIGN} is used for both detector coordinates. The outputs have identical native source occurrences, temporal mapping and weights, assignment residuals, detector-reference slots, source-relation identity, and origin/synthesis state. Numerical validity and payload availability remain coordinate-specific. No relation crosses a Tune state or Tune/readout mapping-revision boundary unless a separately named authority defines that operation, response, and uncertainty; otherwise the affected pair is typed unavailable.

The detector occurrence/time/mapping relation exists independently of a finite numerical x or r payload. AST may therefore construct an otherwise defined coordinate when the exact occurrence, time relation, geometry, aligned observing-state fields, and other required inputs remain valid even if one or both signal coordinates are invalid, unavailable, or not requested. Signal-coordinate facts remain transferred for binding and named downstream policy.

2.3 Native facts and field operators

Each admitted native interface supplies stable observation, interface, row, and occurrence identities; finite ordered time coordinates with named event/epoch meaning; acquisition/integration support; cadence information; direct validity; and timing uncertainty or typed unavailability. Each field also supplies a versioned field identity, unit, unchanged producer frame, topology, missing/validity rule, maximum support span, uncertainty availability, composition history when applicable, and one declared operator:

Continuous scalar	Exact copy at coincidence or bounded adjacent linear interpolation inside one unbroken valid run. Units and frame remain unchanged.
Circular angle	Exact copy or shortest-arc interpolation in a declared period, with typed unavailability at unresolved antipodal ambiguity.
Declared step state	A producer-defined state held only on its explicit half-open validity interval; the held output is synthesized at a non-native time.
Exact-only	Exact coincidence is required. Between samples the field is unavailable.

There is no universal scalar interpolation rule and no extrapolation. Invalid native rows are barriers unless a separate, explicit operator authorizes a different result and preserves its cause. Timestamp and identity coordinates are not interpolated measurements.

2.4 Separated telescope families

The detector-reference occurrence time is the exact grid/time identity t_s; it is not another scalar observing-state field and is never produced by interpolation. The observing-state family describes the science occurrence at that exact time. It includes boresight direction, current elevation, current azimuth where required, and every other declared field needed by AST geometry or field rotation. ALIGN evaluates or maps each producer-declared field at t_s without changing its frame, unit, topology, or meaning. A producer telescope timestamp may remain native support metadata, but it is not a competing current-sample time.

The pointing-correction family is separate. A pointing-support producer selects the exact correction record and native support and supplies record, parent, and selection identities; vector meaning; sign and basis; unit; native support mode and times; covariance availability; and producer semantics. ALIGN supplies only the admitted relation of that selected record to detector-reference time. AST may interpolate the correction only inside the selected native support and never extrapolates. Current science-sample elevation is not inferred from a bracketing correction record.

Optional HWPR fields form a third typed family. Their timing, availability, and alignment may be represented without trimming intensity support. No demodulation, efficiency, calibration, polarization response, or Stokes interpretation follows.

The exact registries for observing-state fields, pointing-correction records, and HWPR fields are unresolved owner/producer decisions. An unresolved entry blocks only the dependent field, correction, coordinate, or polarization claim; ALIGN may not infer a registry from shape, name, numeric range, or storage convention.

2.5 Gaps, interpolation, synthesis, and guards

An acquisition timing gap is a maximal nonempty run of expected but unoccupied native slots inside declared acquisition support. A native occurrence with an invalid or non-finite numerical value remains an original-invalid occurrence, not a timing gap. Ordinary per-field resampling, an acquisition gap, an invalid original, a processing guard, an unavailable optional stream, and a detector-signal continuity surrogate are distinct facts.

A declared per-field interpolation operator does not authorize detector-signal continuity synthesis. A continuity surrogate requires a separately named signal-domain authority and exact count, duration, fraction, edge, causal-support, response, uncertainty, and guard rules. Any authorized surrogate retains exact endpoint rows and weights, causal support, conditional response, uncertainty status, synthesized origin, and zero added acquired exposure. It is never an acquired independent measurement and gains no direct hit, independent statistical weight, degree of freedom, or significance count merely by filling continuity support. The exact surrogate family and limits are unavailable in this v0.1 draft pending `SCI-ALIGN-ODQ-104`; the conditional equation in Section 4 is not authorization.

Physical acquired exposure is native physical integration support attributable to an original occurrence, independent of payload scientific validity. Valid-original exposure is the subset whose original payload and required ALIGN-local mapping facts are valid. Thus an original-invalid occurrence may retain nonzero physical acquired exposure while contributing zero valid-original exposure. Synthesized/surrogate and missing/unoccupied support add zero acquired exposure. Guarded, retained, and use-qualified exposure are later separate facts; they do not rewrite physical acquisition.

Gap identity is observation-wide before chunk slicing and retains its interface, network, detector, or field scope. A cause does not spread to other scopes by convenience. Long-gap observation failure versus marked unavailability is an unresolved owner policy; an unavailable required value always fails closed for the consumer that requires it.

2.6 Five interval identities

The following are separate stable half-open relations:

1. a producer-authoritative physical scan or state interval;
2. a processing chunk;
3. a science window;
4. a context window; and
5. an output selection.

They are not silently merged, renumbered, or used as substitutes. Final partial support, short, empty, guarded, and unusable states retain identity. A processing subset preserves full-observation scan identities. Physical scan segmentation requires producer-authoritative state semantics; the fixed-duration construction in Section 4 is only a conditional processing-window construction and never replaces physical scan authority.

2.7 Exact ALIGN-to-AST transfer

The immutable boundary profile is `SCI-ALIGN_TO_SCI-ASTv0.1/r0.1`. The boundary carries no self-hash; the packet source manifest binds its final bytes by SHA-256. Compatibility requires this exact profile identity, a declared compatible version/revision, and preservation of every required identity, semantic state, ownership boundary, and typed-availability rule. Similar shape or field names do not establish compatibility. A successor names the revision it supersedes and maps every changed, removed, or newly required semantic item.

The transfer is one immutable bundle, not a menu. Every required payload is present with exact identity and semantics or carries typed availability and a reason. It contains:

Family	Required unchanged scientific content
Reference relation	Observation, selected detector-stream reference interface, parent, grid, plan and version, stable ALIGN slot s, cadence, phase, nominal time and cell support, and physical-timing availability.
Native/assigned timing	Native interface and occurrence, acquisition or integration support, corrected native event time when available, nominal slot time, residual, offset-record identity, uncertainty or typed availability, and physical acquired and valid-original exposure as distinct quantities.
Offset record	Producer authority, observation-constant value, positive-add sign convention, seconds unit, reference interface, exactly-once stage/count, validity domain, uncertainty or bound, residual diagnostics, and typed availability.
Source relation	Exact native rows/occurrences, temporal weights, residuals, stable slot identities (o, s), support limits, and mapping-exception identity, expanded or exactly reconstructible from the compact record.
Paired detector relation	Both equations using one A_{ALIGN}, common source-relation and origin/synthesis state, coordinate-specific numerical validity and payload availability, and Tune/mapping-revision boundary.
Observing state	Exact detector-reference occurrence time t_s, plus aligned boresight and observing-state values evaluated or mapped at that exact time, with field and registry identity, unit, unchanged frame, topology, operator, support/source mapping, origin, validity, uncertainty availability, and reason.
Pointing correction	Producer-selected record, parent and selection identities, vector meaning, sign/basis, unit, native support mode/times, covariance availability, and admitted detector-reference time relation.
Origin/support	Original, synthesized, unavailable, original-invalid, and guarded state; continuity availability; source relation; physical acquired exposure, valid-original exposure, zero added surrogate/missing exposure; and separate per-coordinate and per-field validity.
Intervals	Stable physical-scan, processing-chunk, science-window, context-window, and output-selection identities with half-open support and realized state.
Response/uncertainty	Realized operator identity, conditional temporal and nonstationary response or typed availability, propagated covariance when admitted, and timing/model/mapping/selection terms or typed availability.

Family	Required unchanged scientific content
Provenance	Requested, effective, observation-resolved, and realized plan; offset application; registry and operator versions; Tune/readout revision; stable-slot-to-local-row relation for each materialized view; source, exception, and expansion identities; lifecycle; and generation provenance.

AST consumes this occurrence/time/mapping and observing-state bundle without requiring finite detector-signal payload. AST does not reconstruct clocks, choose or repeat ALIGN assignment/interpolation/gap/guard/synthesis policy, infer missing semantics, substitute similarly shaped objects, extrapolate pointing support, collapse local validity into eligibility, or treat nominal slots and synthesized values as physical exposure. No downstream contract reconstructs stable ALIGN slot s from local row j; n is reserved for stable RTC output-sample identity.

2.8 Downstream ownership

AST owns coordinate values, declared frame/topology interpretation, pointing-correction composition, detector-to-coordinate binding, astrometric uncertainty, TAN/WCS astrometric projection, and continuous map coordinates. Target/product authority selects the physical center; AST realizes the named selection without silent recentering. ALIGN has no center authority.

MAP owns sample-to-pixel deposition/gridding, including estimator-specific G_π, kernel, weights, numerical contribution, and map support. The AST TAN/WCS projection is not G_π, and AST coordinate facts do not independently select or authorize it. AST may materialize G_π only for an exact MAP-owned request; ownership remains with MAP.

RTC owns temporal conditioning and propagation of its replacements. CAL owns calibration and atmosphere operations. PTC owns correlated-component estimation and application. VAL evaluates immutable, registered predicates supplied by their named-use owners and does not originate ALIGN facts or policy. Every consumer preserves origin, source support, and zero acquired exposure for synthesized detector values.

3 Assumptions, conditional scope, and typed unavailability

These assumptions are explicit premises, not recovered implementation facts. When a premise is unavailable, only the dependent operation or claim is unavailable or fails closed. No numeric range, storage convention, or model memory supplies a missing premise.

SCI-ALIGN-ASM-001. Every admitted stream supplies stable observation, interface, occurrence, and native-row identities and a strictly ordered time coordinate with producer-authoritative event/epoch meaning.

SCI-ALIGN-ASM-002. Epoch differences and admitted interface offsets can be checked and converted to seconds without guessing. Physical or absolute time claims are conditional on the named producer authority.

SCI-ALIGN-ASM-003. The selected detector-stream interface defines the v0.1 reference grid and clock relation. This selection does not designate a reference detector and does not turn nominal slots into acquisitions.

SCI-ALIGN-ASM-004. V0.1 admits at most one producer-authorized, observation-constant offset per non-reference interface. Drift or state-dependent correction is outside this model and is typed unavailable, not fitted or approximated.

SCI-ALIGN-ASM-005. Tune/readout supplies exact paired $(x, r)^{\mathrm{acq}}$ occurrences and a Tune/mapping-revision identity. Any cross-revision mapping requires separate authority, response, and uncertainty.

SCI-ALIGN-ASM-006. Every aligned producer field has a versioned registry entry that fixes unit, source frame, topology, validity/missing semantics, maximum span, uncertainty availability, and operator before ALIGN applies it.

SCI-ALIGN-ASM-007. A producer-defined state may be held only when its transition event, half-open validity interval, state semantics, and continuity rule are authoritative. Otherwise the field is exact-only.

SCI-ALIGN-ASM-008. A detector-signal continuity surrogate exists only when a named signal-domain authority supplies its family and count, duration, fraction, edge, causal-support, response, uncertainty, and guard rules. Per-field interpolation is not that authority.

SCI-ALIGN-ASM-009. Formal covariance propagation is conditional on supplied joint covariance and the selected payload tier. Timing, model, mapping, and selection uncertainty terms are absent only as typed unavailable, never as numerical zero.

SCI-ALIGN-ASM-010. Physical scans are supplied by an authoritative scan/state relation when they exist. A requested fixed-duration partition is a processing-window policy and cannot replace or rename physical scans.

SCI-ALIGN-ASM-011. All policy parameters are fixed only after requested, effective, and observation-resolved state is recorded. Realized mappings, masks, and intervals are data-derived quantities conditioned upon in the response and covariance equations.

SCI-ALIGN-ASM-012. SCI-ALIGN v0.1 admits only the ordinary nonpolarimetric detector-signal path. The x/r coordinates remain raw KID readout coordinates. Optional HWPR timing/alignment facts neither define nor authorize demodulation, polarization calibration, or Stokes reconstruction.

3.1 Open owner and producer decisions

Identity	Required authority	Typed consequence while unresolved
`SCI-ALIGN-ODQ-101`	Native detector/telescope producers plus SCI-ALIGN	Physical/absolute event-time and offset-validity claims beyond the conditional assigned relation are unavailable for the affected interface.
`SCI-ALIGN-ODQ-102`	Telescope/observing-state producer plus SCI-ALIGN	Coordinate or field-rotation claims needing an unresolved occurrence-local field are unavailable.
`SCI-ALIGN-ODQ-103`	Telescope state producer	Physical scan segmentation or transition claims depending on unresolved `Hold`/state meaning are unavailable.
`SCI-ALIGN-ODQ-104`	SCI-ALIGN gap/continuity authority	Detector-signal synthesis and quantitative gap-limit claims are unavailable; the conditional linear-gap equation is not enabled.
`SCI-ALIGN-ODQ-105`	SCI-ALIGN and SCI-AST interface authorities	Claims requiring an unspecified response, expanded mapping, covariance, model, or selection-uncertainty tier are unavailable.
`SCI-ALIGN-ODQ-109`	HWPR producer plus future polarimetry authority	Unresolved HWPR fields are unavailable, and every HWPR-dependent Stokes claim remains outside v0.1.
`SCI-ALIGN-ODQ-110`	Pointing-support producer plus SCI-ALIGN/SCI-AST	Correction and derived-coordinate claims requiring an unresolved correction-record field are unavailable.

The cause taxonomy in `SCI-ALIGN-ODQ-106`, observing-state/correction family split in `SCI-ALIGN-ODQ-107`, center-selection ownership in `SCI-ALIGN-ODQ-108`, and all other decisions recorded in the exact owner decision register are preserved rather than reopened by these assumptions.

3.2 Claim boundary

The equations and requirements below define a candidate scientific authority. They do not establish that any producer supplies the required facts, that any implementation realizes them, that any result has been validated, or that a system is conformant, frozen, operationally ready, or authorized for production. Such claims require separately scoped evidence and owner review.

4 Canonical equations

The equation identities below are canonical SCI-ALIGN v0.1 author-draft identifiers. Parenthetical aliases preserve the recovered independent-core identities A1–A19. Equations are conditional on the identities, producer semantics, operator registry, and availability rules in Sections 2–3.

4.1 Clock normalization, grid, and assignment

Checked native time on detector-reference coordinate is

$$t_{ik}^{\mathrm{ref}} = (E_i - E_{i_{\mathrm{ref}}})_{\mathrm{sec}} + u_{ik} + \delta_{i\to\mathrm{ref}}, \qquad i_{\mathrm{ref}} = D, \quad \delta_{\mathrm{ref}\to\mathrm{ref}} = 0. \tag{SCI-ALIGN-EQ-001}$$

This preserves recovered-core identity A1. A positive offset is added, acquisition bounds use the same transform, and the operation occurs exactly once before ordering, assignment, gap, interval, or interpolation logic.

For detector phase ϕ_D, cadence $h_D > 0$, and mapped acquisition bounds, the nominal lattice is

$$\begin{aligned} &t_s = \phi_D + s h_D, \\ &s_{\min} = \left\lceil \frac{B_D^- - \phi_D}{h_D} \right\rceil, \qquad s_{\max} = \left\lceil \frac{B_D^+ - \phi_D}{h_D} \right\rceil - 1, \qquad \mathcal{S} = \{s_{\min}, \ldots, s_{\max}\}. \end{aligned} \tag{SCI-ALIGN-EQ-002}$$

This preserves A2. The stable ALIGN slot identity is (o, s). Within a selected materialized view, j is a derived local storage row; it is never external identity and cannot be used to reconstruct s without the explicit view relation. Empty $\mathcal{S}$ is an unsuccessful ALIGN operation. Symbol n is reserved for stable RTC output-sample identity.

An original detector row uses explicit round-half-up assignment:

$$s(k) = \left\lfloor \frac{t_{Dk}^{\mathrm{ref}} - \phi_D}{h_D} + \frac{1}{2} \right\rfloor, \qquad \left| t_{Dk}^{\mathrm{ref}} - t_{s(k)} \right| \le \epsilon_{\mathrm{slot}} < \frac{h_D}{2}. \tag{SCI-ALIGN-EQ-003}$$

This preserves A3. The corrected native time and residual $t_{Dk}^{\mathrm{ref}} - t_{s(k)}$ remain distinct from nominal slot time. Collisions, ambiguity, duplicate/nonmonotonic coordinates, or rows outside declared support fail closed.

4.2 Field operators and paired detector mapping

For a continuous scalar at common time t_s, bounded adjacent interpolation inside one valid run is

$$\begin{aligned} \lambda_s &= \frac{t_s - t_a}{t_b - t_a}, \qquad 0 < \lambda_s < 1, \\ y_s &= (1 - \lambda_s) v_a + \lambda_s v_b. \end{aligned} \tag{SCI-ALIGN-EQ-004}$$

This preserves A4. Exact coincidence is one-hot original support; no extrapolation or skipping across invalid rows is allowed.

For a circular field of declared period P, using signed shortest difference $d_P(a, b) \in [-P/2, P/2)$,

$$y_s = \mathrm{wrap}_P\left(v_a + \lambda_s d_P(v_a, v_b)\right). \tag{SCI-ALIGN-EQ-005}$$

This preserves A5. Exactly antipodal inputs are unavailable unless an explicit unwrap authority resolves the interpolation path; the endpoint choice in the half-open difference interval is not such authority.

The canonical paired detector-coordinate invariant is

$$x^A = A_{\mathrm{ALIGN}} x^{\mathrm{acq}}, \qquad r^A = A_{\mathrm{ALIGN}} r^{\mathrm{acq}}. \tag{SCI-ALIGN-EQ-006}$$

The two applications have the same native occurrences, weights, residuals, slots, source-relation identity, and origin/synthesis state, while numerical validity and payload availability remain coordinate-specific. Equation SCI-ALIGN-EQ-006 does not authorize a mapping across a Tune or mapping-revision boundary.

For one fixed realized mapping and processing window q, a block view is

$$\boldsymbol{y}_q = \begin{bmatrix} S_q G_D & 0 & 0 \\ 0 & S_q I_T & 0 \\ 0 & 0 & S_q I_H \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_D \\ \boldsymbol{v}_T \\ \boldsymbol{v}_H \end{bmatrix} \equiv A_q \boldsymbol{v}. \tag{SCI-ALIGN-EQ-007}$$

This preserves A6. G_D includes the invariant in Equation SCI-ALIGN-EQ-006; absent HWPR is typed unavailable, not zero. Circular mapping is the corresponding nonlinear map and Equation SCI-ALIGN-EQ-007 its local linearization away from ambiguity.

4.3 Conditional continuity surrogate and windows

If and only if a signal-domain authority closes `SCI-ALIGN-ODQ-104`, an authorized internal linear-gap surrogate for g absent slots between valid endpoint values z_0, z_{g+1} may take the form

$$\widehat{z}_\ell = \left(1 - \frac{\ell}{g+1}\right) z_0 + \frac{\ell}{g+1} z_{g+1}, \qquad \ell = 1, \ldots, g. \tag{SCI-ALIGN-EQ-008}$$

This preserves A7 as conditional reasoning, not authorization. Every realized cell retains both endpoint identities and weights, synthesized origin, zero acquired exposure, and conditional response. Leading/trailing, over-limit, invalid-endpoint, or cross-revision gaps are unavailable; no long gap is partially filled.

When a separately requested fixed-duration *processing-window* policy of duration $L > 0$ is admitted, it may resolve to

$$M = \left\lfloor \frac{L}{h_D} + \frac{1}{2} \right\rfloor \geq 1, \qquad \mathcal{S}_q = [s_{\min} + qM, \min\{s_{\min} + (q+1)M, s_{\min} + N\}), \quad q = 0, \ldots, \left\lceil \frac{N}{M} \right\rceil - 1. \tag{SCI-ALIGN-EQ-009}$$

This preserves A8 only for the stated conditional construction. It does not define, replace, or renumber producer-authoritative physical scans. Final partial and short windows retain identity.

The corresponding selector is

$$(S_q)_{as} = \mathbf{1}\{s = s_{\min} + qM + a\}, \qquad s \in \mathcal{S}_q. \tag{SCI-ALIGN-EQ-010}$$

This preserves A9. Slicing changes support and window response but not units, slot identity, physical-scan identity, or origin.

4.4 Response and uncertainty

For fixed A_q and temporal template $\boldsymbol{p}$, the conditional response is

$$R_q(\boldsymbol{p}) = \frac{\partial \mathbb{E}[\boldsymbol{y}_q \mid A_q]}{\partial \alpha} = A_q \boldsymbol{p}, \qquad \mathbb{E}[\boldsymbol{v}] = \alpha \boldsymbol{p} + \boldsymbol{\mu}_0. \tag{SCI-ALIGN-EQ-011}$$

This preserves A10. It is conditional ALIGN response, not a downstream estimator response.

For uniform samples and fractional target $(k+\alpha)h$, $0 \leq \alpha \leq 1$, linear interpolation relative to an ideal fractional delay has response

$$H_\alpha(\omega) = e^{-i\omega\alpha h} \left[(1-\alpha) + \alpha e^{i\omega h}\right]. \tag{SCI-ALIGN-EQ-012}$$

This preserves A11. $H_\alpha(0) = 1$; amplitude and phase are not unity in general. Gap response is row-specific and nonstationary.

For fixed affine mapping and supplied full joint native covariance,

$$\begin{aligned} \mathbb{E}[\boldsymbol{y}_q \mid A_q] &= A_q \mathbb{E}[\boldsymbol{v} \mid A_q], \\ C_{y_q|A_q} &= A_q C_{v|A_q} A_q^{\mathsf{T}}. \end{aligned} \tag{SCI-ALIGN-EQ-013}$$

This preserves A12, including known cross-time, cross-detector, and cross-stream terms.

For Equation SCI-ALIGN-EQ-004, one scalar variance is

$$\operatorname{Var}(y_s) = (1-\lambda_s)^2 \operatorname{Var}(v_a) + \lambda_s^2 \operatorname{Var}(v_b) + 2\lambda_s(1-\lambda_s) \operatorname{Cov}(v_a, v_b). \tag{SCI-ALIGN-EQ-014}$$

This preserves A13. Shared endpoints induce output covariance; smoother appearance or smaller diagonal variance is not new information.

For differentiable signals and timing parameter vector $\boldsymbol{\eta}$, the first-order timing term is

$$C_{\text{timing}} = J_\eta C_\eta J_\eta^{\mathsf{T}}, \qquad (J_\eta)_{ab} = \frac{\partial y_a}{\partial \eta_b}. \tag{SCI-ALIGN-EQ-015}$$

This preserves A14. It is systematic/response uncertainty, not detector noise.

The full reporting target is

$$C_y = C_{y|A} + C_{\text{timing}} + C_{\text{interp.model}} + C_{\text{policy/selection}}. \tag{SCI-ALIGN-EQ-016}$$

This preserves A15. Missing terms are unavailable, not zero; conditional claims name what was conditioned upon.

For twice differentiable scalar truth on a bracket of width $\Delta = t_b - t_a$,

$$|f(t) - If(t)| \le \frac{\Delta^2}{8} \sup_{s\in[t_a,t_b]} |f''(s)|. \qquad \text{(SCI-ALIGN-EQ-017)}$$

This preserves A16 and is available only with a scientifically justified curvature bound.

4.5 Exposure and local state

For nominal cell I_s and detector occurrence or stable detector identity d, physical acquired exposure and its valid-original subset are

$$e^{\mathrm{acq}}_{sd} = \begin{cases} |I_s \cap I^{\mathrm{native}}_{kd}|, & \text{one original physical detector occurrence } k, \\ 0, & \text{synthesized, missing, or unoccupied support,} \end{cases} \qquad \text{(SCI-ALIGN-EQ-018)}$$
$$e^{\mathrm{vo}}_{sd} = e^{\mathrm{acq}}_{sd}\mathbf{1}\{\text{original payload and required ALIGN-local mapping facts valid}\}.$$

This preserves A17 while separating physical acquisition from scientific usability. An original-invalid payload may have $e^{\mathrm{acq}}_{sd} > 0$ and $e^{\mathrm{vo}}_{sd} = 0$. Synthesized/surrogate and missing/unoccupied support add no acquired exposure. Nominal duration, elapsed span, guarded/retained/use-qualified exposure, and downstream weight remain separate later facts and never rewrite physical acquisition.

Each output cell has the product state

$$\mathcal{S}_{\mathrm{ALIGN}} = (\text{origin}, \text{local validity}, \text{method}, \text{reason}, \text{source IDs}, \text{weights}). \qquad \text{(SCI-ALIGN-EQ-019)}$$

This preserves A18, specialized so that local validity is not AST validity or downstream eligibility.

For a factual ALIGN-local relation requiring an original valid detector value and aligned field set $\mathcal{V}$,

$$m^{\mathrm{local}}_{sd} = \mathbf{1}\{D_{sd} \text{ is original and locally valid}\} \prod_{v\in\mathcal{V}} \mathbf{1}\{T_{vs} \text{ is available and locally valid}\}. \qquad \text{(SCI-ALIGN-EQ-020)}$$

This preserves recovered-core A19 only as an ALIGN-local mapping/support fact. It is not a universal eligibility bit, AST validity, or authorization for an estimator. A named-use owner supplies any additional predicate, and its designated evaluator applies every applicable restriction.

5 Normative requirements

"Must" and "shall" below are normative for this candidate authority. The stable identifiers are sequential and must not be reassigned; a successor may supersede a requirement only through an explicit semantic mapping.

5.1 Boundary, identity, and paired detector relation

SCI-ALIGN-REQ-001. SCI-ALIGN shall operate after Tune/readout has transformed native $(I, Q)^{\mathrm{acq}}$ into exact native paired $(x, r)^{\mathrm{acq}}$ occurrences and before AST, RTC, CAL, PTC, VAL, MAP, or other scientific conditioning.

SCI-ALIGN-REQ-002. "Detector-reference" shall identify the selected detector-stream reference interface, admitted clock relation, and assigned grid, never a reference detector.

SCI-ALIGN-REQ-003. Every admitted input and output shall retain stable observation, parent, interface, occurrence, native-row where applicable, detector, plan, grid, stable ALIGN slot s, and version identities sufficient to distinguish identity from row position or numerical time. Stable ALIGN identity is (o, s); j is local storage row only, and no downstream contract shall reconstruct s from j. Symbol n is reserved for stable RTC output-sample identity.

SCI-ALIGN-REQ-004. ALIGN shall realize Equation SCI-ALIGN-EQ-006 with identical native source occurrences, temporal weights, residuals, slot identities, source-relation identity, and origin/synthesis state for x and r, while preserving coordinate-specific numerical validity and payload availability.

SCI-ALIGN-REQ-005. ALIGN shall not align I, Q, mix x with r, fill one coordinate from the other, or cross a Tune state or Tune/readout mapping-revision boundary without separately named operation, response, and uncertainty authority.

SCI-ALIGN-REQ-006. The exact detector occurrence/time/mapping relation shall remain available independently of whether either numerical detector-coordinate payload is finite, valid, available, or requested.

5.2 Clock relation and detector-reference grid

SCI-ALIGN-REQ-007. Every native interface shall supply a finite strictly ordered time coordinate with named event/epoch meaning, unit, acquisition or integration support, cadence information, validity, and uncertainty or typed unavailability.

SCI-ALIGN-REQ-008. Epoch normalization and each admitted interface offset shall follow Equation SCI-ALIGN-EQ-001; a positive offset is added to place the native coordinate later on detector-reference time.

SCI-ALIGN-REQ-009. Each non-reference offset record shall identify producer authority, observation-constant value, sign, seconds unit, reference interface, application stage and count, validity domain, uncertainty or bound, residual diagnostics, and typed availability.

SCI-ALIGN-REQ-010. An admitted interface offset shall be applied exactly once, before ordering checks, slot assignment, gap identity, interval construction, or interpolation, and shall never be rounded to an integer number of samples.

SCI-ALIGN-REQ-011. Absent authority, incomparable epochs, ambiguous sign/unit/reference or stage, drift, or state dependence outside the declared constant-offset domain shall make the affected physical-time or mapping claim unavailable or unsuccessful; ALIGN shall not fit a time-varying correction.

SCI-ALIGN-REQ-012. The selected detector interface shall define the lattice in Equation SCI-ALIGN-EQ-002; detector acquisition support shall not be trimmed merely because telescope or optional auxiliary support is missing.

SCI-ALIGN-REQ-013. Original detector rows shall use the round-half-up assignment and declared tolerance in Equation SCI-ALIGN-EQ-003, with $0 \leq \epsilon_{\text{slot}} < h_D/2$.

SCI-ALIGN-REQ-014. Nonpositive cadence, empty detector support, nonmonotonic or duplicate native time identity, ambiguous assignment, two original rows in one slot, or a row outside declared acquisition support shall make the ALIGN operation unsuccessful before scientific conditioning.

SCI-ALIGN-REQ-015. Nominal assigned slot time, corrected native event time, assignment residual, native acquisition/integration support, nominal cell support, and acquired exposure shall remain distinct and queryable.

5.3 Fields and telescope-family separation

SCI-ALIGN-REQ-016. Every aligned field shall have a versioned registry identity, unit, unchanged source frame, topology, validity and missing rule, maximum support span, uncertainty availability, producer semantics, and one exact operator class before it is mapped.

SCI-ALIGN-REQ-017. A continuous scalar may use only exact copy or bounded adjacent interpolation inside one unbroken valid run; endpoints are exact, invalid rows are barriers, and extrapolation is prohibited.

SCI-ALIGN-REQ-018. A circular field shall declare its period and use the exact shortest signed difference interval $[-P/2, P/2)$ in Equation SCI-ALIGN-EQ-005; an exactly antipodal interpolation shall be typed unavailable absent explicit unwrap authority. A bounded non-wrapping angle may use scalar interpolation only under its declared topology.

SCI-ALIGN-REQ-019. A categorical, Boolean, counter, mode, flag, or hardware state shall never be linearly interpolated. It may be held only under producer-authorized half-open state semantics; otherwise it is exact-only.

SCI-ALIGN-REQ-020. The detector-reference occurrence time t_s shall be an exact grid/time identity, not an interpolated observing-state field. ALIGN shall evaluate or map the declared boresight, elevation, required azimuth, and other occurrence-local observing-state fields at that exact time while preserving their producer meaning, units, frames, topology, support, and causes. A producer telescope timestamp may remain native metadata but shall not become a competing current-sample time.

SCI-ALIGN-REQ-021. Producer-selected pointing-correction records shall remain a separate family with exact record, parent, selection, vector-meaning, sign/basis, unit, native-support, time, covariance-availability, and producer identities; ALIGN shall supply only their admitted detector-reference time relation.

SCI-ALIGN-REQ-022. ALIGN shall not infer current science-sample elevation or other occurrence-local observing state from a pointing-correction record. AST may interpolate a producer-selected correction only within its native support and shall never extrapolate it.

SCI-ALIGN-REQ-023. Optional HWPR fields shall retain their own versioned identities, event semantics, units, frames, topology, operator, support, availability, and reason. Missing HWPR shall not trim ordinary nonpolarimetric detector-signal support.

5.4 Gaps, synthesis, exposure, and intervals

SCI-ALIGN-REQ-024. Acquisition timing gaps, invalid original rows, ordinary field resampling, processing guards, unavailable optional streams, and detector-signal continuity synthesis shall be represented as distinct causes and shall retain their interface/network/detector/field scope.

SCI-ALIGN-REQ-025. A declared per-field interpolation operator shall not authorize a detector-signal continuity surrogate. Each surrogate requires a separately named signal-domain authority and exact count, duration, fraction, edge, causal-support, response, uncertainty, and guard rules.

SCI-ALIGN-REQ-026. Until `SCI-ALIGN-ODQ-104` is decided, quantitative detector surrogate and gap-limit claims shall remain unavailable; the conditional form in Equation SCI-ALIGN-EQ-008 shall not be enabled by analogy or convenience.

SCI-ALIGN-REQ-027. Every authorized synthesized detector value shall retain exact source rows and weights, causal support, mapping identity, conditional response, uncertainty status, synthesized origin, and zero added acquired exposure.

SCI-ALIGN-REQ-028. A synthesized detector value shall never be relabeled as an original independent occurrence and shall gain no direct hit, independent statistical weight, degree of freedom, or significance count merely through continuity support.

SCI-ALIGN-REQ-029. Leading or trailing gaps shall not be extrapolated; over-limit or invalid-endpoint gaps shall not be partially filled; and each maximal gap shall retain stable half-open identity, extent, classification, action, and realized count.

SCI-ALIGN-REQ-030. Nominal duration, elapsed span, physical acquired exposure, valid-original exposure, synthesized/surrogate support, missing/unoccupied support, guarded/retained/use-qualified exposure, continuity availability, and downstream weight shall be separately reported according to Equation SCI-ALIGN-EQ-018. Payload invalidity or later policy shall not rewrite physical acquisition.

SCI-ALIGN-REQ-031. Physical-scan, processing-chunk, science-window, context-window, and output-selection identities shall remain distinct stable half-open intervals and shall not be silently merged, renumbered, or substituted.

SCI-ALIGN-REQ-032. Producer-authoritative physical scan/state relations shall take precedence over processing-window construction. Unresolved producer `Hold`/transition semantics shall block only dependent physical-scan claims.

SCI-ALIGN-REQ-033. If a fixed-duration processing-window policy is separately admitted, it shall use the recorded rounding and partition in Equation SCI-ALIGN-EQ-009, preserve stable full-observation identities, retain final partial, short, empty, and unusable states, and reject overlap unless a successor defines multiple membership and exposure.

5.5 Validity, response, uncertainty, and provenance

SCI-ALIGN-REQ-034. Origin, method, reason, source identities, weights, payload availability, finiteness, producer validity, ALIGN mapping validity, support, uncertainty availability, physical acquired exposure, and valid-original exposure shall be separately representable; a numeric sentinel shall not be the sole encoding.

SCI-ALIGN-REQ-035. Detector-occurrence identity, signal-coordinate validity, ALIGN mapping validity, AST coordinate validity, and downstream use-specific eligibility shall remain five distinct causes. None shall alias or rescue another.

SCI-ALIGN-REQ-036. ALIGN shall export factual local mapping validity and support and shall not author a global science-eligibility bit. A named-use owner shall author its predicate and a designated evaluator shall apply every applicable restriction.

SCI-ALIGN-REQ-037. For every realized mapping, ALIGN shall provide an exact conditional response or typed response availability sufficient for the admitted detail tier. DC preservation shall not be generalized to time-varying unity response.

SCI-ALIGN-REQ-038. When admitted input covariance is supplied and the selected tier supports propagation, ALIGN shall use Equation SCI-ALIGN-EQ-013 and retain known cross-time, cross-detector, and cross-stream terms; it shall not invent independent white noise, inverse variance, or a dense covariance.

SCI-ALIGN-REQ-039. Timing, interpolation-model, mapping, and policy/selection uncertainty terms shall be reported when admitted or as typed unavailable with reason. Unavailable shall never be encoded or interpreted as zero.

SCI-ALIGN-REQ-040. Requested, effective, observation-resolved, and realized state shall be distinct and flow one way, with exact accepted values, normalized values, observation facts, applied mappings, exceptions, diagnostics, and reasons recorded at their owning stage.

SCI-ALIGN-REQ-041. State shall be owned by one reduction request and replaced per observation. Sequential repeated runs and every admitted alternate or parallel path shall produce the same identities, mappings, states, ordering, and diagnostics within the declared numerical gate, without state leakage.

SCI-ALIGN-REQ-042. The standard persisted relation shall be compact and generative: grid, offset, registry, interval, mapping/operator, aggregate support, and exception identities shall exactly reconstruct ordinary identity and every nonidentity state.

SCI-ALIGN-REQ-043. Expanded rows, endpoints, weights, residuals, selected covariance, and forensic details may be requested by tier only when their availability is explicit and their omission does not change semantics, numerical results, or the exact reconstructibility of the always-on compact relation.

SCI-ALIGN-REQ-044. Failure to produce required mapping, state, availability, or realized provenance shall propagate as an unsuccessful operation; logging and continuing as success is prohibited.

5.6 AST transfer and ownership restrictions

SCI-ALIGN-REQ-045. ALIGN shall export the one immutable `SCI-ALIGN_TO_SCI-ASTv0.1/r0.1` transfer bundle described in Section 2.7; similar shape, field names, or separately substituted objects shall not establish compatibility.

SCI-ALIGN-REQ-046. Every required AST-transfer payload shall be present with exact identity and semantics or typed availability and reason, including reference grid, timing, offset, source relation, paired detector facts, observing state, pointing correction, origin/support, intervals, response/uncertainty, and complete provenance.

SCI-ALIGN-REQ-047. AST shall consume the immutable occurrence/time/mapping and aligned observing-state relation without reconstructing clocks; choosing, replacing, or repeating ALIGN policy; inferring missing semantics; extrapolating pointing support; or requiring finite numerical x/r merely to bind an otherwise defined coordinate.

SCI-ALIGN-REQ-048. AST shall own geometry, field rotation, pointing-correction composition, detector-coordinate binding, astrometric uncertainty, TAN/WCS astrometric projection, and continuous map coordinates; target/product authority shall own physical-center selection.

SCI-ALIGN-REQ-049. MAP shall own sample-to-pixel deposition/gridding projection, estimator-specific G_π, kernel, weights, contribution, and map support. AST may materialize G_π only for an exact MAP-owned request, without transferring ownership.

SCI-ALIGN-REQ-050. RTC, CAL, PTC, VAL, MAP, and every admitted consumer shall preserve ALIGN identities, origin, support, response, covariance availability, and exposure facts and shall own only their named transformation or policy.

SCI-ALIGN-REQ-051. SCI-ALIGN v0.1 shall admit only the ordinary nonpolarimetric detector-signal path. The x/r coordinates remain raw KID readout coordinates. Optional HWPR timing or alignment shall not authorize polarization demodulation, efficiency, calibration, response, or Stokes reconstruction.

SCI-ALIGN-REQ-052. SCI-ALIGN shall not infer any unavailable producer event, epoch, field, topology, transition, unit, frame, clock, support, correction, uncertainty, or policy meaning from plausible values, implementation conventions, or similarly shaped data.

SCI-ALIGN-REQ-053. All internal sample, slot, and scan indices shall be zero-based and all internal intervals half-open. A persisted one-based scan number shall be a named representation adapter and shall not alter internal identity.

SCI-ALIGN-REQ-054. The output shall include detailed facts and aggregate counts or exposure together with an availability manifest that is internally consistent with the compact generative mapping and every exception.

SCI-ALIGN-REQ-055. No SCI-ALIGN output or this authority shall by itself claim implementation conformity, observational validation, scientific approval, freeze, readiness, production authorization, or an end-to-end downstream response.

6 Edge cases and falsifiable predictions

These predictions are pre-observation, implementation-independent comparison targets; listing them is not validation. Categorical identities, counts, memberships, source relations, and masks compare exactly. For deterministic double-precision fixtures, finite scalar results compare at relative tolerance 10^{-12} and absolute tolerance $64\epsilon_{\text{mach}}\max(1, h_D)$ in the declared unit. Circular values compare by wrapped distance at 10^{-12} rad. Available covariance compares at relative tolerance 10^{-12} with exact sparsity and source-weight identity. A lower-precision successor must declare and justify its tolerance before examining results.

Predictions 001–018 preserve recovered-core T01–T18 in order. Predictions 019–026 append falsifiers required by the approved paired-coordinate, AST boundary, cause-separation, telescope-family, and compact-detail decisions.

SCI-ALIGN-PRED-001. For telescope native times $0, 1, 2$ s, values $0, 1, 2$, $\delta_{T\to\text{ref}} = +0.25$ s with $i_{\text{ref}} = D$, and target $t = 1$ s, corrected source times shall be $0.25, 1.25, 2.25$ s and the aligned value 0.75, with weights $0.25, 0.75$; no integer rounding is allowed. Reversing the offset sign shall fail this value. (Recovered-core T01.)

SCI-ALIGN-PRED-002. Exact coincidence and valid endpoints shall produce one-hot weights, unchanged value, unit, and frame, and original origin. Targets before the first or after the last admitted coordinate shall be unavailable, not extrapolated. (T02.)

SCI-ALIGN-PRED-003. Constant and affine continuous-scalar fixtures on coincident and fractional grids shall reproduce analytically, and constant/DC response shall be one. (T03.)

SCI-ALIGN-PRED-004. Sinusoidal injections at multiple fractional positions and frequencies, including near Nyquist, shall match complex amplitude and phase from Equation SCI-ALIGN-EQ-012; no unity response away from DC may be claimed. (T04.)

SCI-ALIGN-PRED-005. Using shortest signed difference in $[-P/2, P/2)$, a declared circular pair $359°, 1°$ at midpoint shall wrap to $0°$, not $180°$. An exactly antipodal pair shall be unavailable absent explicit unwrap state. (T05.)

SCI-ALIGN-PRED-006. Categorical and step fields shall reject linear interpolation. A producer-declared step state shall use its half-open hold interval and have synthesized/held provenance at a non-native target. (T06.)

SCI-ALIGN-PRED-007. Conditional on owner authorization under `SCI-ALIGN-ODQ-104`, one-, two-, and maximum-admitted-length internal detector gaps in constant and ramp fixtures shall form one maximal run and match Equation SCI-ALIGN-EQ-008 with exact source identities, weights, synthesized state, and zero acquired exposure. Until authorization, the synthesis result shall be unavailable. (T07 specialized.)

SCI-ALIGN-PRED-008. Leading, trailing, over-limit, invalid-endpoint, and cross-revision gaps shall never be extrapolated or partially filled; their unavailable reason shall be exact, and a native non-finite row shall remain original-invalid rather than become a timing gap. (T08 specialized.)

SCI-ALIGN-PRED-009. Within-tolerance detector jitter shall retain exact assignment residuals. A half-slot ambiguity admitted only by an invalid tolerance, duplicate/nonmonotonic timestamp identity, or two rows assigned to one slot shall fail before conditioning. (T09.)

SCI-ALIGN-PRED-010. For the separately admitted fixed-duration processing-window case $N = 7, M = 3$, windows shall be $[0, 3), [3, 6), [6, 7)$; the final sample is retained, an exact boundary has one membership, and indices remain zero-based. This fixture makes no physical-scan claim. (T10 specialized.)

SCI-ALIGN-PRED-011. Short and empty interval identities shall be recorded without padding or renumbering; a subset preserves full-observation identities; overlap is rejected absent successor authority. (T11.)

SCI-ALIGN-PRED-012. Two interpolated outputs sharing correlated source endpoints shall reproduce full ACA^{T}, Equation SCI-ALIGN-EQ-014, and nonzero cross-output covariance; smoothness shall not create independent information. (T12.)

SCI-ALIGN-PRED-013. With supplied nonzero offset covariance on a known ramp, the timing term shall match Equation SCI-ALIGN-EQ-015. With a scientifically justified curvature bound, a quadratic fixture shall satisfy Equation SCI-ALIGN-EQ-017; without those inputs the respective terms shall be unavailable, not zero. (T13.)

SCI-ALIGN-PRED-014. For valid-original, original-invalid, synthesized, missing/unoccupied, guarded, retained, use-qualified, and window-sliced cells, physical acquired exposure shall follow original physical integration support, valid-original exposure shall be its valid subset and sum once, and synthesized or missing support shall add zero acquired exposure. An original-invalid cell may have nonzero physical acquired but zero valid-original exposure; later guard/use state shall not rewrite it. Nominal and elapsed span remain distinct. (T14.)

SCI-ALIGN-PRED-015. Missing telescope support shall not trim the detector grid; affected telescope fields shall be unavailable. Absent optional HWPR shall be explicit and shall authorize no polarization output. (T15.)

SCI-ALIGN-PRED-016. Two observations and repeated runs with distinct offsets, gaps, fields, and intervals shall reconstruct independent requested, effective, observation-resolved, and realized state with no leakage. (T16.)

SCI-ALIGN-PRED-017. For every admitted alternate or parallel execution path over Predictions 001–016, integer, state, identity, mapping, provenance, and ordering outputs shall match exactly and finite values shall meet the same declared tolerance. A required path skip is not a pass. (T17.)

SCI-ALIGN-PRED-018. Injected failure of required mapping, state, availability, provenance, or output publication shall make the operation unsuccessful; a log-and-continue success shall falsify the contract. (T18.)

SCI-ALIGN-PRED-019. Given distinct finite x^{acq} and r^{acq} payloads on the same occurrences, both outputs shall have identical source rows, temporal weights, residuals, stable slots s, source-relation identity, and origin/synthesis state while retaining their distinct numerical values and coordinate-specific validity. Any cross-mixing or different mapping falsifies Equation SCI-ALIGN-EQ-006.

SCI-ALIGN-PRED-020. If the only candidate interpolation bracket crosses a Tune state or Tune/readout mapping-revision boundary and no separate authority is provided, both affected paired outputs shall be typed unavailable across that boundary; neither coordinate shall be synthesized or used to rescue the other.

SCI-ALIGN-PRED-021. In a factorial fixture varying detector occurrence identity, x/r validity, ALIGN mapping validity, AST input validity, and named-use predicate independently, the five causes shall remain separately queryable. Invalid x/r alone shall not invalidate an otherwise defined coordinate; an available coordinate shall not rescue invalid ALIGN mapping or authorize a named downstream use.

SCI-ALIGN-PRED-022. When aligned current elevation evaluated at exact occurrence time t_s differs numerically from a field carried in a selected pointing-correction record, the transfer shall retain both families, identities, meanings, and supports separately. Current field-rotation state shall not be inferred from the correction record, and a correction target outside selected native support shall be unavailable rather than extrapolated.

SCI-ALIGN-PRED-023. An exact `SCI-ALIGN_TO_SCI-ASTv0.1/r0.1` bundle shall be accepted only with every required identity, semantics, and typed-availability rule. A byte-mutated boundary or similarly shaped payload without a declared compatible profile/revision and semantic mapping shall not be treated as compatible.

SCI-ALIGN-PRED-024. For identical inputs and resolved state, compact-only and expanded-detail output tiers shall reconstruct the same mapping, exceptions, conditional response, identities, states, counts, exposures, and numerical results. Requested detail shall not change science.

SCI-ALIGN-PRED-025. Removing one unresolved optional field-registry entry shall make only that field and its dependent claim unavailable. It shall not trim the detector relation, fabricate semantics, invalidate an unrelated AST input, or enable a downstream eligibility bit.

SCI-ALIGN-PRED-026. A physical-scan identity and a processing chunk with equal numerical bounds shall remain distinct parents. Subsetting or repartitioning chunks shall preserve physical-scan, science/context-window, and output selection identities and shall not double-count exposure.

6.1 Fail-closed edge conditions

The following conditions are unsuccessful before scientific conditioning or typed unavailable for the exact dependent output: invalid or ambiguous offset unit/sign/reference/stage; incomparable epoch; nonpositive cadence; empty detector support; nonmonotonic or duplicate time identity; ambiguous or colliding slot assignment; undeclared field topology/operator; invalid interval overlap; missing required observing-state support; unauthorized signal synthesis; cross-Tune-revision mapping; extrapolated pointing support; or inability to publish required realized provenance. Missing optional HWPR is nonfatal for the ordinary nonpolarimetric detector-signal path while every polarimetric interpretation remains unavailable.

These predictions do not pre-authorize fixtures, claim that any executable path exists, or report a pass. They become validation evidence only through a separately governed comparison against a named implementation and exact artifacts.

Assessment boundary

The requirements and predictions are durable comparison targets. No comparison was performed in this Stage B targeted revision, and no statement here establishes implementation behavior, observational performance, scientific conformity, validation success, operational readiness, or authorization for production.